INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

STRAINERS, SEDIMENT, PIPELINE, SELF-CLEANING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers self-cleaning pipeline sediment strainers.

1.1 Intended use. The strainers specified are intended for installation in fuel piping systems on the discharge side of the pump.

2. SALIENT CHARACTERISTICS.

2.1 Codes and standards. Strainers shall comply with the applicable requirements of table I.

Table I. Reference data.

Equipment/Characteristic	Description
Fuel type	F-76 fuel in accordance with MIL-DTL-16884
Operating pressure	600 psi (max)
Temperature range	70 to 120 °F
Piping connections: Inlet and outlet Sump drain	 3-inch, Class 600, ASME B16.5 raised face flanges 1-inch, Class 600, ASME B16.5
Screen	0.007 slots equivalent or better than 80-mesh per linear inch
Pressure resistance	FCI 78-1
Flow capacity	80 gpm in accordance with ISA-75.02
Shock resistance	MIL-S-901, Grade A
Vibration resistance	MIL-STD-167-1, Type I

2.2 Design and construction. Each strainer shall contain one rotatable screen and floating brush for the purpose of cleaning the screen without shutting down operation of the system. The sump below the screen shall be fitted with a valved drain outlet.

2.3 Material. The material requirements for all strainer components shall be in accordance with table II.

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FSC 4730

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Table II. Materials required for strainers.

Part	Material	Applicable Documents	Remarks
Housing	CRES	ASTM A351/A351M	CF8M
Cover	Monel		N04400
Brush	Stainless steel	ASTM A240/A240M	S30400
Threaded fasteners	Non-CRES Non-ferrous	NASM17829	
		ASTM F593	
O-rings	Fluorocarbon	SAE-AS3209	Type I, Class 1
Screen	CRES plate	ASTM A240/A240M	S31600

2.4 Screen. The screen shall be made of one unit. The screen shall be of the spiral wound "V" shape wire construction. All seam welds shall be continuous.

2.4.1 Screen strength test. The strainer screen shall be tested by applying a differential pressure across the screen equal to 100 percent of the specified strainer design pressure using water for a duration of 10 minutes. Differential pressure shall be applied in a rapid manner for 1 minute or less. Screens that distort, stretch, or exhibit other weaknesses shall be rejected.

2.5 Actuation. Each strainer screen shall be fitted with a shaft along its rotational axis, which penetrates the compartment cover through a watertight seal. The design shall incorporate a positive means of preventing stem ejection. The hand wheel shall be sized and configured so that maximum required operating force is less than 50 pounds.

2.6 Brush. The brush shall be aligned so that the screen can be turned in either direction without being damaged.

2.7 Covers. Compartment covers shall be flanged and attached to the body by bolts. Each cover shall be equipped with an anti-spray feature to deflect fluid spray in the event of an O-ring seal failure.

2.8 Envelope dimensions. Envelope dimensions shall be in accordance with table III.

Table III. Envelope dimensions.

Strainer Size	Width	Flange-to-Flange	Height	Screen Removal Length
3	10.00	14.75	18.25	8.50 (max)
NOTES: 1. Height does not include drain valve. 2. All dimensions are in inches.				

2.9 Functional performance. Unless otherwise specified (see 6.1), design flow capacity and design temperature shall be as specified in table I.

2.9.1 Operational cycling tests. The strainer shall be tested for a total of 600 cycles at design pressure. At the completion of the 600 cycles, compliance with the required operating force of 50 pounds or less and zero leakage shall be demonstrated. Upon satisfactory completion of this test, the unit shall be refurbished to new condition before being offered for delivery.

2.10 Workmanship. The strainer body and covers shall be of uniform quality and condition, free from blow holes, porosity, hard spots, shrinkage defects, cracks, and other defects. All surfaces shall be smooth and well cleaned. The inside surfaces of strainers shall be well cleaned and free from sharp edges.

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3. REGULATORY REQUIREMENTS.

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 Prohibited material. Zinc or cadmium plating shall not be used on any part of the strainer.

3.3 Mercury exclusion. The strainer shall be free of mercury contamination. During the manufacturing process, tests, and examination, the product to be offered for acceptance shall not come in direct contact with mercury or any of its components, nor with any mercury containing device employing a single boundary of containment.

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same production offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Additional inspection. Unless otherwise specified in the contract or order (see 6.1), where other specifications form a part of this CID, sampling, examination, and tests shall be conducted as required by the pertinent specification.

5. PACKAGING.

5.1 General requirement. Commercial packaging shall be in accordance with the requirements of ASTM D3951.

5.2 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage, and storage.

6. NOTES.

6.1 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. The required design flow capacity and design temperature if not as specified (see 2.9 and table I).
c. Product conformance provisions (see 4.2).
d. Whether repair parts are required (see 6.3.1).

6.2 Sources of documents.

6.2.1 Aerospace Industries Association (AIA). The NASM standard is available from Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3901 or online at www.aia-aerospace.org.

6.2.2 American Society of Mechanical Engineers (ASME). ASME standards are available from the American Society of Mechanical Engineers, 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.

6.2.3 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

6.2.4 Defense specifications and standards. Defense specifications and standards are available online at http://quicksearch.dla.mil.

6.2.5 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

6.2.6 Fluid Control Institute (FCI). FCI standards are available from the Fluid Controls Institute, Inc., 1300 Sumner Avenue, Cleveland, OH 44115 or online at http://www.fluidcontrolsinstitute.org/.

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6.2.7 International Society of Automation (ISA). ISA standards are available from the International Society of Automation, 67 T.W. Alexander Dr., P.O. Box 12277, Research Triangle Park, NC 27709 or online at www.isa.org.

6.2.8 SAE International. SAE standards are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.

6.3 Ordering spare or repair parts. When ordering spare parts or repair parts for the equipment covered by this CID, the contract should state that such spare parts and repair parts should meet the same requirements as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.3.1 Repair parts and repair parts boxes. Complete sets of repair screens for each type and class should be furnished as follows:

a. One set for each of the eight initial strainers required.
b. One set for each two additional strainers required.
c. Cover O-rings - four for every strainer supplied.
d. Other O-rings and seals - one set for every eight strainers supplied.

When specified (see 6.1), items 6.3.1.c and 6.3.1.d should be furnished in repair boxes.

6.4 Preinstallation instruction. A set of instructions covering the preinstallation of the equipment should be furnished. Instructions should include all information necessary to return the unit to active status, such as, but not limited to, the addition of lubricants prior to operation, flushing of lines, removal of greaseproof barrier, and the location of detached components. Instructions should be packaged in a transparent waterproof plastic bag. Closure should be by heat sealing. The shipping container in which the instructions are packed should be so marked.

MILITARY INTERESTS:

Custodians:
Army – MR
Navy – SH
Air Force – 99

Review Activities:
Navy – SA
Air Force – 71
DLA – CC

Preparing Activity:
Navy – SH
(Project 4730-2017-033)

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